MIL-L-46675B
2 September 1976
SUPERSEDING
MIL-L-46675A
22 September 1972

MILITARY SPECIFICATION

LINK, CARTRIDGE: METALLIC BELT, CALIBER .30, M1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of metallic belt link intended to hold caliber .30 ammunition in the form of flexible belts for use in caliber .30 machine guns.

* 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	-	Preservation-Packaging, Methods of
MIL-W-13855	-	Weapons: Small Arms and Aircraft Armament Subsystems, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Federal

Fed. Std. No. 356	-	Commercial Packaging of Supplies and Equipment
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions

DRAWINGS

Rock Island Arsenal

D5559722	-	Link, Cartridge, Metallic Belt, Caliber .30, M1

FSC 1305

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PUBLICATIONS

Rock Island Arsenal

P5559722 - Packaging Data Sheet for Link, Cartridge, Metallic Belt, Cal. .30, M1

IEL5559722 - Index of Inspection Equipment Lists, Link, Metallic Belt Cal. 30, M1

(Acceptance Inspection Drawings referenced in this specification form a part of IEL5559722.)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Material (ASTM)

ASTM E18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1) the first article shall include a pilot pack (see 5.1).

3.2 Material, construction and design. The links shall conform to the material, construction and design requirements specified herein, on Drawing D5559722 and in MIL-W-13855.

3.2.1 Elongation. Links shall withstand a tensile load of 35 + 0.5 pounds and the total elongation of the pitch dimension after removal of the load shall not exceed 0.008 inch permanent set. Testing shall be as specified in 4.5.3.5.

3.2.2 Cartridge extraction. The force required to extract a steel plug of minimum cartridge size from the links shall be not less than 5 pounds nor more than 10 pounds. The loading depth of the plug shall be 2.248 inch + .030 inch from the butt end of the plug to the farthest extremity of the neck ring. Testing shall be as specified in 4.5.3.6.

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3.2.3 Assembleability and free hinging. Links shall assemble and disassemble with caliber .30 Government standard dummy ammunition without scratching or shaving resulting in the removal of metal from the cartridges. When assembled into 100 round belts, links shall hinge without binding, grabbing or freezing. Testing shall be as specified in 4.5.3.7.

3.2.4 Functioning. Links shall be capable of being assembled with Government standard caliber .30 ball cartridges into 100 round belts and fired in an approved caliber .30 machine gun without cracking, breaking, separating or malfunctioning. Testing shall be as specified in 4.4.

3.3 Marking. Links shall be clearly marked in accordance with MIL-W-13855.

3.4 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.1). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein, the functioning test specified in 4.6.8 and such other inspection as necessary to determine that all requirements of the contract have been met.

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* 4.4.1 Examination and test. Five components from the first article shall be inspected for all unlisted drawing dimensions and tests.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified herein, the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105 and MIL-W-13855. Samples shall be withdrawn in accordance with MIL-STD-105 from packaged links. Inspection lots shall not exceed 500,000 links per lot.

4.5.1.1 Heat treat lot. The heat treat lot shall be limited to one heat of steel and shall consist of links heat treated in a batch type furnace at one charge or in a continuous type furnace during an uninterrupted period extending over not more than one eight hour shift.

4.5.2 Examination. Examination of links shall be as specified herein and in MIL-W-13855.

4.5.2.1 Classification of defects. Classification of defects and acceptable quality levels (AQL's) for examination in accordance with MIL-STD-105 shall be as specified herein and shall include the following provisions:

a. Unless otherwise specified, AQL's apply to each listed characteristic.
b. Where a Government drawing is listed as the method of inspection, the contractor shall use inspection equipment fabricated in accordance with the applicable drawing. In the event the contractor desires to use alternate equipment which equals or exceeds the accuracy requirements of the specified drawing, it shall be submitted to the Government for approval.
c. Where Visual is specified as the method of inspection of protective coating, the coating shall be visually examined for completeness, uniformity in appearance, in color, for freedom from pits, corrosion, scratches, and worn or bare spots.
d. Where Visual is specified as the method of dimensional inspection, the sample shall be either scaled or compared with a specimen of known acceptable quality.
* e. Inspection for Workmanship shall include examination for missing manufacturing operations and the presence of nicks, burs, cracks, scratches, tool marks, deformations or other signs of poor workmanship.
* f. Examination for packaging defects specified in 4.5.2.1.2 shall apply to each link or container, as applicable, using inspection Level I.

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4.5.2.1.1 Link, Metallic Belt, Caliber .30, M1, Drawing D5559722.

Categories	Characteristic	AQL	Inspection Method
Critical:	None defined		
Major:			
101	Cracks, laminations, tears and deformations.	0.65	Visual
102	Center distance between rings.	0.65	6516340
103	Inside diameter, connecting ring.	0.65	7127962
104	Inside diameter, neck ring.	0.65	7127963
105	Angular location, neck flare.	0.65	7184648
106	Offset of bridge.	0.65	7190109
107	Height of bridge wings.	0.65	7190113
108	Distortion limits.	0.65	6057626
Minor:			
201	Missing or defective protective coating.	1.5	Visual
202	Missing or improper manufacturers identification marking.	1.5	Visual
203	Width of connecting loop.	1.5	7190118
204	Workmanship (3.4).	4.0	Visual

4.5.2.1.2 Packaging. Unless otherwise specified in each listed characteristic, the packaging requirements are specified on Packaging Data Sheet P5559722. The inspection method shall be visual.

Categories	Characteristic	AQL
Major:		
101	Improper level of preservation and packing (see procurement documents).	1.0
102	Incorrect or illegible marking.	1.0
103	Contaminated links.	1.5
104	Inadequate cleaning and drying.	1.5
105	Improper containers.	1.5
106	Container interior contaminated with dirt, lint or other foreign matter.	1.5
107	Incorrect quantity of links per shipping container.	1.5
108	Improperly oriented links in cartons.	1.5
109	Improperly oriented cartons in shipping container.	1.5
110	Missing VCI treated sheets between rows of unit packs (applicable to level A only).	1.5

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Categories	Characteristic	AQL
Major:		
111	Improper closure of shipping container.	1.5
112	Incorrect or missing inspection lot number on each carton of links.	1.5
Minor:		
201	Workmanship.	4.0

4.5.3 Testing.

4.5.3.1 Certification. Unless otherwise specified, the contractor shall furnish the Government representative with certified statements that each inspection lot conforms to the materials and processes specified on the applicable drawings and specifications.

4.5.3.2 Protective finish testing.

* 4.5.3.2.1 Accelerated corrosion testing. Sample links shall be selected from inspection lots for each processing bath prior to application of supplementary wax coating. Lot formation and sampling shall be in accordance with MIL-P-16232, whether manual basket dip or continuous conveyor equipment is used. The sample links shall be tested for compliance with the requirements of MIL-P-16232 using the test method specified in 4.6.1.1.

4.5.3.2.2 Supplementary wax coating weight testing. A sample of 10 links with supplementary wax coating shall be taken from each shift's production from each production machine and tested for supplementary wax coating weight using the test method specified in 4.6.1.2. Failure of links to meet the specified supplementary wax coating weight requirements shall be cause for rejection of the represented lot.

* 4.5.3.3 Hardness testing. A sample of 32 links shall be taken from each heat treat lot and tested for hardness using the test method specified in 4.6.2. Failure of any link shall be cause for rejection of the heat treat lot.

4.5.3.4 Decarburization testing. A sample of 5 links shall be selected from each heat treat lot and tested for decarburization using the test method specified in 4.6.3. Failure of any link shall be cause for rejection of the heat treat lot.

* 4.5.3.5 Elongation testing. A sample of 32 links shall be taken from each heat treat lot and tested for elongation (see 3.2.1) using the test method specified in 4.6.4. Failure of any link shall be cause for rejection of the heat treat lot.

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* 4.5.3.6 Cartridge extraction testing. A sample of 200 links shall be taken from each inspection lot and tested for cartridge extraction (see 3.2.2) using the test method specified in 4.6.5. Failure of three or more links shall be cause for rejection of the inspection lot.

4.5.3.7 Assembleability and free hinging testing. A separate sample of 800 links shall be selected from each packaged inspection lot and tested for the free hinging and assembleability requirements (see 3.2.3) using test methods specified in 4.6.6. Failure of any link shall be cause for rejection of the lot.

4.5.3.8 Determination of cleanliness testing. Links shall be tested for cleanliness using the sampling procedure and test method specified in 4.6.7.

4.5.4 Inspection equipment.

4.5.4.1 Acquisition, maintenance and disposition. Unless otherwise specified, (see 6.1) responsibility for acquisition, maintenance and disposition of acceptance inspection and test equipment prescribed on the List of Inspection Equipment IEL5559722, and for all other inspection equipment required by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6 Test methods.

4.6.1 Protective finish tests.

* 4.6.1.1 Accelerated corrosion test. Sample links shall be tested for corrosion resistance of the applied protective coating using either the salt spray (fog) test or Springfield Immersion test specified in MIL-P-16232.

* 4.6.1.2 Supplementary wax coating weight test. The sample links shall be tested by weighing the links together to the nearest milligram. The supplementary wax coating shall then be completely removed from the links by trichloroethylene or trichlorethane vapor degreasing, and links dried and reweighed. The difference between the total weights before and after removal of the supplementary wax coating divided by the number of links tested shall be considered as the weight of the supplementary wax coating of each link.

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* 4.6.2 Hardness test. The sample links shall be tested for hardness requirements using ASTM E18 and fixture D7190100.

4.6.3 Decarburization test. The cross section of the sample links shall be polished, etched with a 2 to 5 percent nital solution and examined under a microscope of not less than 500 power to determine compliance with decarburization requirements on Drawing D5559722.

4.6.4 Elongation test. The sample links shall be tested using inspection equipment in accordance with Drawing D7190140. Links shall be placed on the mandrels of the fixture, the indicator gage set to zero, and the load specified applied. The load shall then be removed from the link and the indicator gage used to determine whether compliance has been met. At the completion of this test the sample links shall be scrapped.

4.6.5 Cartridge extraction test. The sample links shall be tested using the inspection equipment specified on Drawing F6516202. Links shall be placed in the holding fixture and the steel plug inserted in the link to the depth of 2.248 inches + .030 inch. The load shall then be gradually applied until the plug is extracted from the link.

* 4.6.6 Assembleability and free hinging test. Sample links shall be tested by assembling the links into a 100 round belt using Government standard caliber .30 dummy cartridges. The belt shall then be laid out horizontally to its full length and one end drawn back and forth over the remainder of the belt to test for free hinging. After testing for free hinging, the belt shall be disassembled. During assembly and disassembly operations the cartridges shall be visually examined for evidence of scratching or shaving resulting in the removal of metal.

4.6.7 Determination of cleanliness test. The links shall be tested for cleanliness requirements using MIL-P-116.

4.6.8 Functioning test. A sample of 100 links taken from the first article shall be tested by the testing agency for functioning requirement (see 3.2.4) by assembling into a 100 round belt with Government standard caliber .30 ball cartridges and firing the belt in an approved caliber .30 machine gun. Firing shall be in bursts of 10 to 20 rounds each.

5. PACKAGING

5.1 Pilot pack. A pilot pack shall consist of links preserved in accordance with Packaging Data Sheet P5559722 for the level of protection specified in the contract (see 6.1), packed level B and forwarded as specified in the contract (see 3.1).

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5.2 Levels A and B. Preservation, packing and marking shall be in accordance with Packaging Data Sheet P5559722 for the level of protection specified in the contract (see 6.1).

5.3 Commercial packaging. Preservation, packing and marking shall be in accordance with Fed. Std. No. 356.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements:

a. Title, number and date of this specification.
b. List of drawings, publications and specifications pertinent to the link, showing applicable revision dates.
c. Requirements for submission of first article and pilot pack (see 3.1, 4.4 and 5.1).
d. Responsibilities for acquisition, maintenance and disposition of acceptance inspection equipment, if other than specified (see 4.5.4.1).
e. Responsibility for furnishing dummy cartridges.
f. Selection of applicable levels of protection (A or B), or commercial packaging (5.1, 5.2 and 5.3).

6.1.2 Contract data requirements. When warranted, requirements for reports of the results of examination and testing should be specified on a DD Form 1423 included in the contract.

6.2 Management control system requirements. When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208 on the Management Control Systems Summary List, DD Form 1660.

6.3 Unless otherwise specified (see 6.1) the contract should specify the application of MIL-I-45607 on the Management Control Systems Summary List, DD Form 1660.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing Activity:
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DOCUMENT IDENTIFIER AND TITLE

MIL-L-46675B; Link, Cartridge: Metallic Belt, Caliber .30, M1

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

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